Light

GCSE Computer Science — Cyber Security Threats (Set 1)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A company's computers become infected with software designed to cause damage.
What is the general term for this type of harmful software?

- **A.** Firmware
- **B.** Malware
- **C.** Freeware
- **D.** Shareware

Q2 (1 mark)

A type of malware attaches itself to legitimate programs and spreads when the infected program is run.
What type of malware is this?

- ○ **A.** A worm
- ○ **B.** A firewall
- ○ **C.** A virus
- ○ **D.** An encryption key

Q3 (1 mark)

A type of malware spreads automatically across networks without any user action.
What type of malware is this?

- ○ **A.** A worm
- ○ **B.** A Trojan
- ○ **C.** A keylogger
- ○ **D.** Ransomware

Q4 (1 mark)

A user downloads what appears to be a free game, but it secretly installs harmful software.
What type of malware is this?

- ○ **A.** A worm
- ○ **B.** A virus
- ○ **C.** Spyware
- ○ **D.** A Trojan

Q5 (1 mark)

A user's files become encrypted and a message demands payment for the decryption key.
What type of malware has infected the computer?

- ○ **A.** Ransomware
- ○ **B.** A keylogger
- ○ **C.** A worm
- ○ **D.** A virus

Q6 (1 mark)

A type of malware secretly records every key pressed on a keyboard.
What is this malware called?

- ○ **A.** Ransomware
- ○ **B.** A firewall
- ○ **C.** A Trojan
- ○ **D.** A keylogger

Q7 (1 mark)

An email pretending to be from a bank asks a user to click a link and enter their login details.
What type of attack is this?

- ○ **A.** A brute force attack
- ○ **B.** A phishing attack

- ○ **C.** A denial of service attack
- ○ **D.** A SQL injection attack

Q8 (1 mark)

An attacker manipulates people into revealing confidential information rather than exploiting technical vulnerabilities.
What is this type of attack called?

- ○ **A.** Network scanning
- ○ **B.** Data mining
- ○ **C.** Social engineering
- ○ **D.** Packet sniffing

Q9 (1 mark)

Software monitors incoming and outgoing network traffic and blocks unauthorised connections.
What is this software called?

- ○ **A.** A firewall
- ○ **B.** A keylogger
- ○ **C.** A compiler
- ○ **D.** An operating system

Q10 (1 mark)

Software scans files and programs to detect and remove malicious software.
What type of software is this?

- ○ **A.** Anti-malware software
- ○ **B.** A word processor
- ○ **C.** A spreadsheet application
- ○ **D.** A web browser

Q11 (1 mark)

Data is converted into a coded format that can only be read with the correct key.
What is this process called?

- ○ **A.** Compression
- ○ **B.** Encryption
- ○ **C.** Defragmentation
- ○ **D.** Authentication

Q12 (1 mark)

A company creates copies of important files and stores them in a separate location.
What is this practice called?

- ○ **A.** Encryption
- ○ **B.** Defragmentation
- ○ **C.** Backup

○ **D.** Compression

Q13 (2 marks)

A student compares computer viruses and worms.
Which statement correctly describes a key difference between them?

○ **A.** Viruses spread automatically across networks; worms require user action to spread

○ **B.** Viruses require a host program and user action to spread; worms spread automatically without user intervention

○ **C.** Viruses encrypt files for ransom; worms record keystrokes

○ **D.** Viruses only affect mobile devices; worms only affect desktop computers

Q14 (2 marks)

A hospital's computer systems are infected with ransomware.
Which combination of impacts would this attack cause?

○ **A.** Faster computer performance and increased storage space

○ **B.** All computers automatically upgrade to the latest operating system

○ **C.** Patient records become inaccessible and the hospital may need to pay criminals to restore access

○ **D.** Network speed increases and security improves

Q15 (2 marks)

A keylogger has been installed on a victim's computer.
How would an attacker use this to steal the victim's bank account credentials?

- ○ **A.** By encrypting all the victim's files until they reveal their password
- ○ **B.** By redirecting the victim to a fake banking website
- ○ **C.** By blocking the victim's internet connection until they pay a ransom
- ○ **D.** By recording every keystroke when the victim types their username and password on the banking website

Q16 (2 marks)

An attacker phones a company employee, pretending to be from the IT department, and asks for their password to 'fix a problem'.
What type of attack is this, and why is it effective?

- ○ **A.** A social engineering attack; it is effective because it exploits human trust rather than technical vulnerabilities
- ○ **B.** A brute force attack; it is effective because it tries all possible passwords
- ○ **C.** A denial of service attack; it is effective because it overwhelms the network
- ○ **D.** A SQL injection attack; it is effective because it exploits database weaknesses

Q17 (2 marks)

A company configures its firewall to block incoming connections on certain ports.
How does this improve network security?

- ○ **A.** It speeds up all network traffic by reducing congestion
- ○ **B.** It automatically encrypts all data transmitted through the network
- ○ **C.** It prevents attackers from accessing services through those blocked ports while allowing legitimate traffic through permitted ports
- ○ **D.** It removes all malware from incoming email attachments

Q18 (2 marks)

A company implements access control so that employees can only access files relevant to their job role.
Why is this an important security measure?

- ○ **A.** It makes files load faster for all employees
- ○ **B.** It automatically backs up all company files
- ○ **C.** It encrypts all files on the network
- ○ **D.** It limits the damage if an employee's account is compromised and reduces insider threat risks

Q19 (2 marks)

A school creates an acceptable use policy (AUP) for students using school computers.
What is the main purpose of this policy?

- ○ **A.** To define rules for appropriate computer use and inform users of their responsibilities and consequences of misuse
- ○ **B.** To make computers run faster by limiting the software installed
- ○ **C.** To provide free internet access to all students

- ○ **D.** To automatically block all websites except educational ones

Q20 (2 marks)

A company hires security experts to deliberately attempt to break into their systems to find weaknesses.
What is this practice called?

- ○ **A.** Social engineering
- ○ **B.** Penetration testing
- ○ **C.** Phishing
- ○ **D.** Data mining

Q21 (2 marks)

A computer has not installed the latest security updates from the software manufacturer.
Why does this create a security risk?

- ○ **A.** The computer will run faster without updates
- ○ **B.** Known vulnerabilities remain unfixed and can be exploited by attackers
- ○ **C.** The computer will have more storage space available
- ○ **D.** Updates always contain malware

Q22 (2 marks)

A company employs an ethical hacker to test their security.
How does ethical hacking differ from malicious hacking?

- ○ **A.** Ethical hackers use more advanced tools than malicious hackers
- ○ **B.** Ethical hackers only test websites; malicious hackers only test networks
- ○ **C.** Ethical hackers have permission and aim to improve security; malicious hackers act illegally to cause harm or steal data
- ○ **D.** There is no difference; both types of hacking are illegal

Q23 (2 marks)

A server room has keycard access, CCTV cameras, and is kept locked at all times.
What type of network security measures are these?

- ○ **A.** Software security measures
- ○ **B.** Encryption measures
- ○ **C.** Backup measures
- ○ **D.** Physical security measures

Q24 (3 marks)

A business experiences the following incident:
• An employee receives an email appearing to be from their bank
• They click a link and enter their password on a fake website
• Malware is downloaded that records all future keystrokes
• The attacker uses captured credentials to access company systems
Which combination of attack types was used in this incident?

- ○ **A.** Denial of service, followed by SQL injection, then ransomware
- ○ **B.** Brute force attack, followed by worm infection, then data encryption

○ **C.** Phishing to obtain initial credentials, followed by Trojan delivery of a keylogger, enabling further credential theft

○ **D.** Physical theft of hardware, followed by password guessing

Q25 (3 marks)

A company's servers are infected with ransomware. The attackers demand £50,000 in Bitcoin for the decryption key.
The IT manager proposes three response options:
• Option A: Pay the ransom immediately to restore operations
• Option B: Restore from the most recent backup (made yesterday)
• Option C: Attempt to decrypt files without paying using security tools
Which evaluation of these options is correct?

○ **A.** Option A is best because paying guarantees data recovery and is the fastest solution

○ **B.** Option B is best because it restores data without funding criminals, though some recent work may be lost; Option C may work but is unreliable; Option A funds criminal activity and offers no guarantee of receiving the key

○ **C.** Option C is best because all ransomware can be easily decrypted with free tools

○ **D.** All options are equally effective and there is no significant difference between them

Q26 (3 marks)

A company wants to reduce the risk of successful social engineering attacks against its employees.
Which combination of measures would be most effective?

○ **A.** Installing faster computers, upgrading to the latest operating system, and increasing internet bandwidth

○ **B.** Blocking all email communication, disconnecting from the internet, and removing all telephones

○ **C.** Increasing password complexity requirements, installing additional firewalls, and encrypting all files

○ **D.** Regular security awareness training, verification procedures for sensitive requests, and simulated phishing exercises to test employee responses

Q27 (3 marks)

A school implements the following security measures:
• Firewall blocking unnecessary ports
• Anti-malware software on all computers
• Regular software updates applied automatically
• User accounts with minimum necessary permissions
• Daily encrypted backups stored off-site
Which statement correctly evaluates this security approach?

○ **A.** This approach is flawed because it uses too many different security measures

○ **B.** Only the firewall is necessary; the other measures are redundant and wasteful

○ **C.** This approach is ineffective because it does not include any physical security measures

○ **D.** This is a defence-in-depth approach using multiple layers of protection, so if one measure fails or is bypassed, others remain to protect the system

Q28 (3 marks)

A penetration tester reports these findings to a company:
• Web server running software with known vulnerabilities (last updated 18 months ago)

• Several employee accounts using 'password123' as their password
• USB ports enabled on all computers in public areas
• No monitoring of failed login attempts
Which combination correctly categorises these as technical or human vulnerabilities?

- ○ **A.** All four are technical vulnerabilities because they involve computer systems
- ○ **B.** All four are human vulnerabilities because humans configured the systems
- ○ **C.** Weak passwords are the only vulnerability; the others are normal system configurations
- ○ **D.** Outdated software and enabled USB ports are technical vulnerabilities; weak passwords and lack of monitoring are human/procedural vulnerabilities

Q29 (3 marks)

A company's IT audit reveals:
• Anti-malware software installed on all computers
• Virus definitions last updated 8 months ago
• Operating system security patches applied monthly
• Firewall configured and active
A new ransomware variant released 3 months ago infects several computers.
Which statement correctly explains why the infection occurred despite having security measures?

- ○ **A.** The infection occurred because firewalls cannot block any malware
- ○ **B.** The outdated virus definitions meant the anti-malware could not recognise the new ransomware variant, which was created after the last update
- ○ **C.** The infection occurred because operating system patches were applied too frequently

○ **D.** The infection was inevitable because no security measures can prevent ransomware

Q30 (3 marks)

A company implements a backup strategy with:
• Full backup every Sunday night
• Incremental backups every weekday night
• Backups stored on a network drive in the same building
• No encryption of backup files
On Thursday, ransomware encrypts all company files including the backup drive.
Which statement correctly evaluates the weaknesses in this backup strategy?

○ **A.** The strategy failed because full backups should be performed daily, not weekly

○ **B.** The strategy failed because incremental backups are less reliable than full backups

○ **C.** The backup location (same network) allowed ransomware to encrypt backups too; off-site or offline storage would have protected them; lack of encryption also means stolen backups could be read

○ **D.** The strategy was perfect; ransomware would have encrypted any backup regardless of location

Total: **55** marks • Time: **30 min**

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